



FINAL RESEARCH PROJECT

Cracking the Code of Employee Turnover: A Python-Based Investigation

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Course:

CIS 2266 - Python Data Analytics

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The Attrition Problem & Data Context



Strategic Context

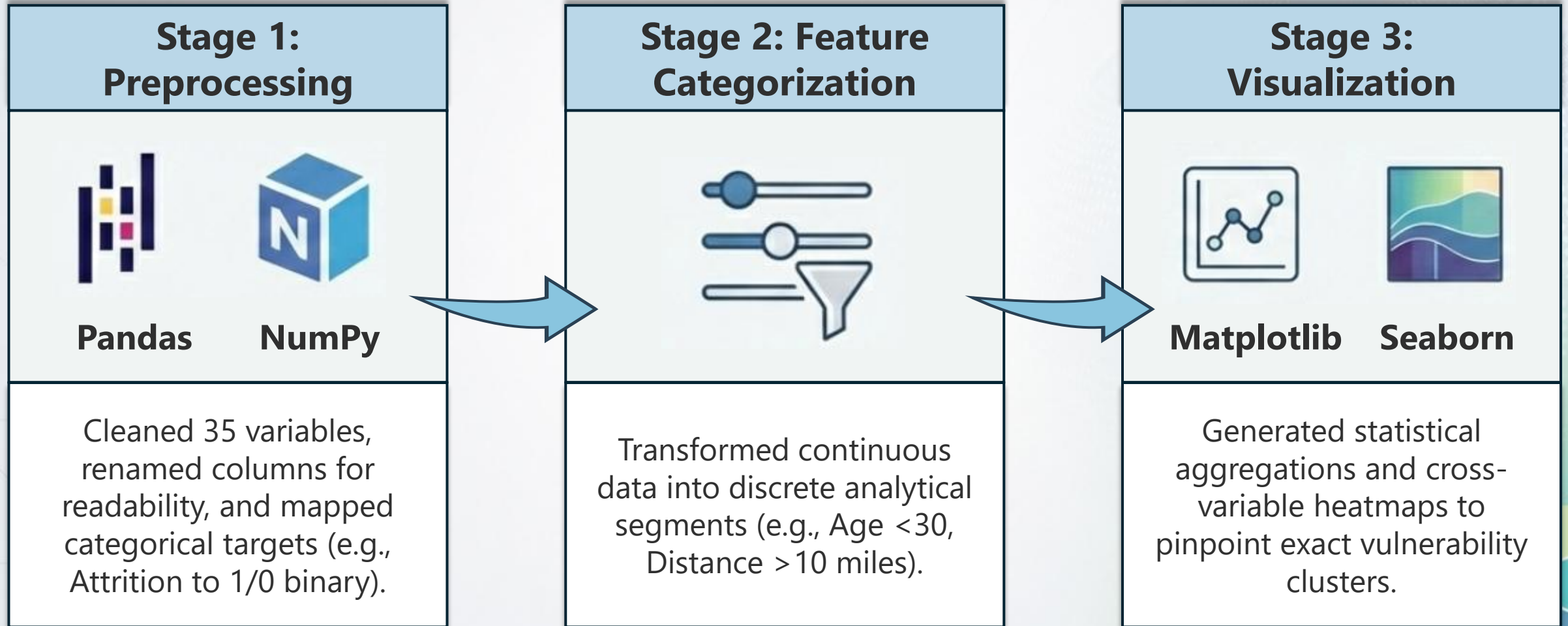
- **Impact:** Employee turnover significantly impacts organizational stability and increases recruitment costs.
- **Focus:** Move beyond simple metrics to identify the specific "breaking points" where workload exceeds compensation.
- **Goal:** Provide HR with a data-driven roadmap to stabilize high-risk employee segments.



Analytical Foundation

- **Source:** High-fidelity synthetic records from IBM's HR Research division (via Kaggle)
- **Scale:** **1,470** detailed employee profiles ensuring statistical significance.
- **Breadth:** **35** diverse features including compensation, commute strain, and marital status.
- **Processing:** Cleaned and analyzed using Python (Pandas/Seaborn).

The Python Analytics Pipeline



RQ1: Pay vs. Workplace Stress



High Workload Intensity

30.5%

Attrition rate when **Overtime** is required (triples the baseline).

24.9%

Attrition rate under **Frequent Business Travel**.

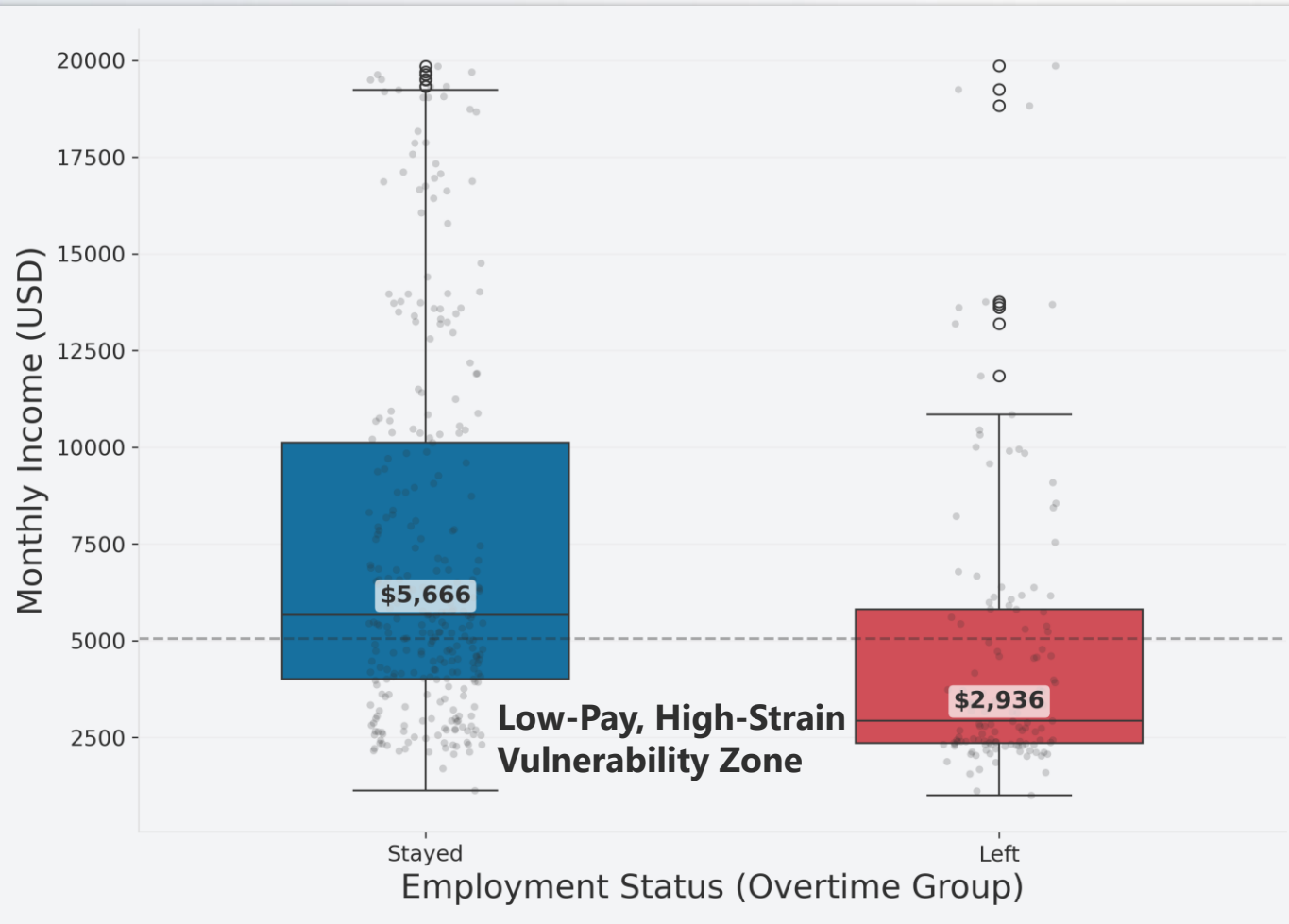


The Moderating Hypothesis

*"Does **Monthly Income** act as a primary moderating variable against this workload burnout?"*

Method: Isolated high-stress employees to compare the median income of those who stayed versus those who left.

The \$5,000 Retention Threshold



Median Income Gap: Retained employees consistently earn more despite identical overtime conditions.

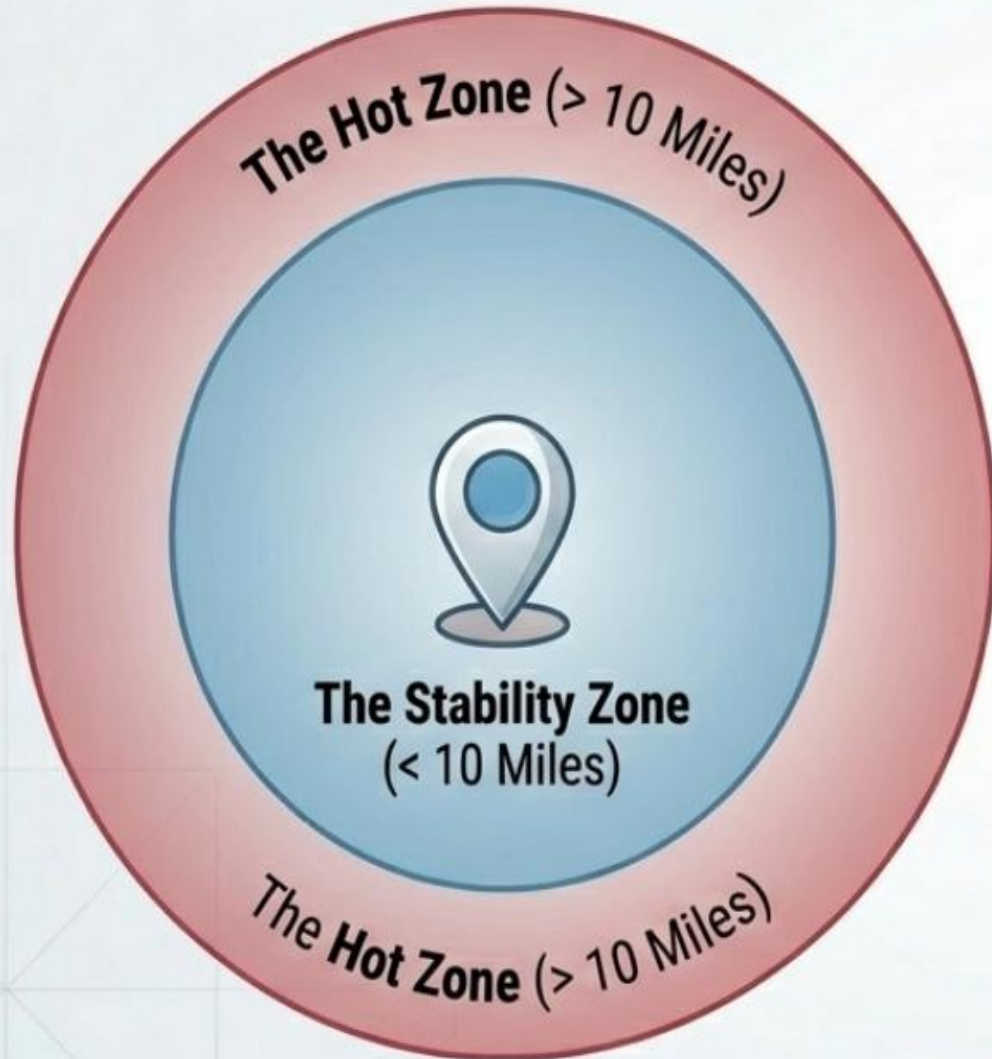


The Buffer Effect: Attrition is heavily concentrated below the \$5,000/month threshold.



High-Income Resilience: Specialized roles with premium pay easily tolerate extended hours.

RQ2: Geography vs. Stability



The Commute Catalyst

Distance from home accelerates turnover, but its impact is entirely dependent on an employee's life stage.



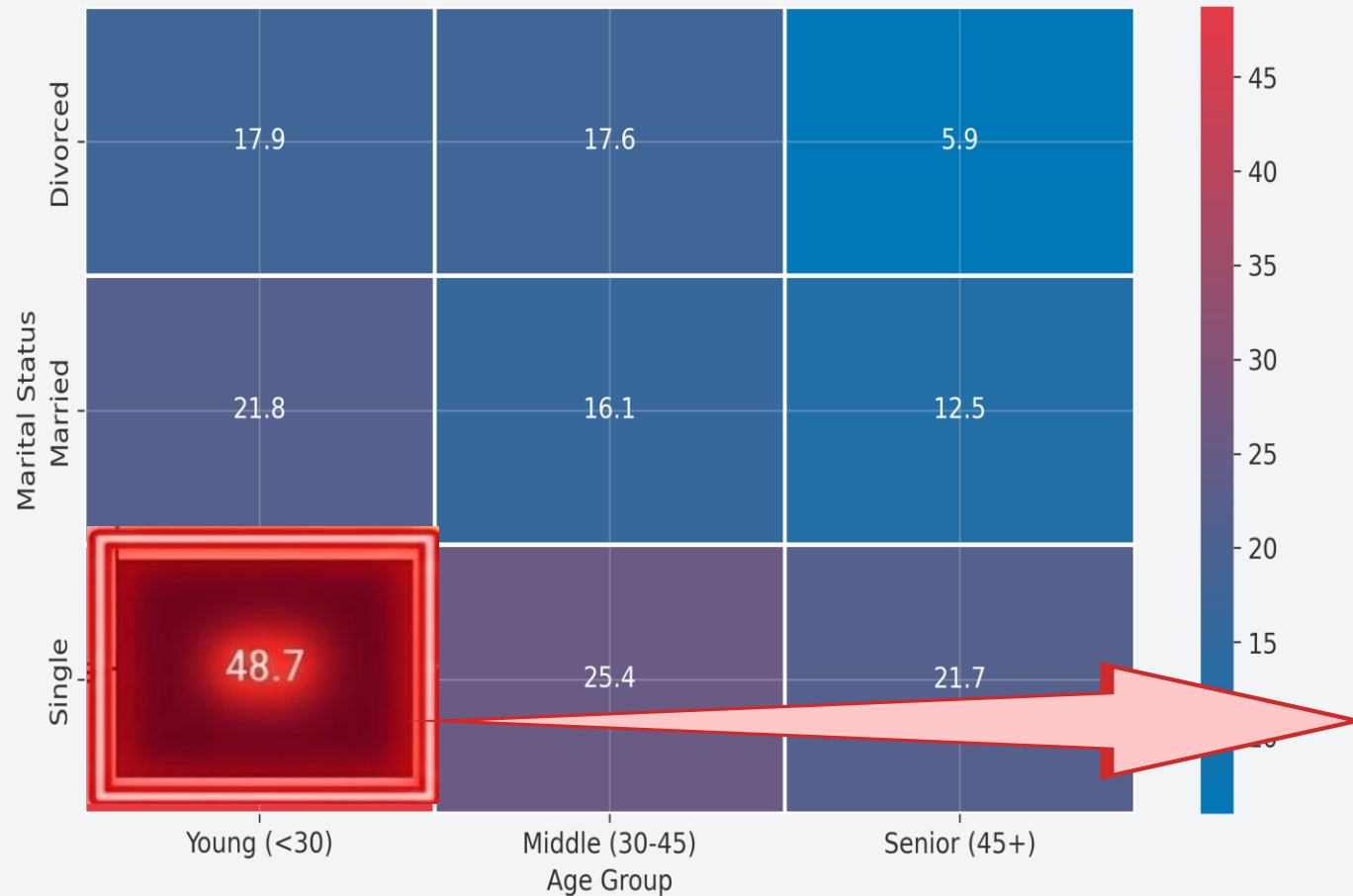
Stability Zone (<10 miles): Retention remains universally high across all demographics.



Hot Zone (>10 miles): Risk magnifies drastically (often doubling) for young and single professionals.

Pinpointing Maximum Attrition Risk

Attrition Rate (%) - Far Commute Group (>10 miles)



Maximum Volatility: Young, Single employees face a massive **48.7%** attrition rate when commuting more than **10 miles**.



The Demographic Buffer: Senior (45+) & Married employees show high resilience; attrition drops to **5.9%**.



Conclusion: Domestic stability and career maturity neutralize geographic strain.

Strategic Recommendations

 Identified Risk	 Targeted HR Response
Risk 1: Income-Stress Gap Roles under \$5,000/month with mandatory overtime.	Targeted Compensation Adjustments: Implement targeted retention bonuses for high-travel/overtime roles below the \$5,000 threshold.
Risk 2: Demographic Strain Single staff <30 living 10+ miles away (48.7% Risk).	Precision Flexibility: Prioritize hybrid/remote work and commute subsidies for the young demographic in the 'Hot Zone'.

Conclusion: Data-driven HR strategies allow for surgical interventions that stabilize the workforce and optimize budgets.